



Department of Mathematics and Natural Sciences

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HomeWork-1

(Answer all questions. Limit and Continuity, each question contains 5 marks, $20 \times 5 = 100$. Differentiability, each question contains 10, $10 \times 5 = 50$. Total : $100 + 50 = 150$)

Limit:

1. $\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + 3x} + 2x)$
2. $\lim_{x \rightarrow \infty} x - \ln x$
3. $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$
4. $\lim_{x \rightarrow 0} \frac{\ln(x+1)}{\cos x + e^x - 1}$
5. $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^2}}$
6. $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{7x}\right)^{3x+5}$

$$7. \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

$$8. \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}}$$

$$9. f(x) = \begin{cases} \frac{1}{x+2} & , x < -2 \\ x^2 - 5 & , -2 < x < 3 \\ \sqrt{x+13} & , x > 3 \end{cases} \text{ find } \lim_{x \rightarrow 2} f(x) \text{ and } \lim_{x \rightarrow 3} f(x)?$$

$$10. f(x) = \begin{cases} x^2 + 1 & , x < 0 \\ 1 & , x = 0 \\ 1 + x & , x > 0 \end{cases} \text{ find } \lim_{x \rightarrow 0} f(x) ?$$

Continuity:

$$1. f(x) = \begin{cases} 4 - \frac{1}{2}x, & x < 2 \\ \sqrt{x+c}, & x \geq 2 \end{cases}, \text{ find the value of } c \text{ such that limit exists and } f(x) \text{ is continuous at } x = 2.$$

$$2. f(x) = \begin{cases} cx^3 + 2x, & x < 2 \\ x^3 - cx, & x \geq 2 \end{cases}, \text{ find the value of } c \text{ such that limit exists and } f(x) \text{ is continuous at } x = 2.$$

$$3. \text{ Show that } f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x^2}\right), & x \neq 0 \\ 0 & , x = 0 \end{cases} \text{ is continuous at } x = 0.$$

$$4. \text{ Show that } f(x) = |x+1| + |x| \text{ is continuous at } x = -1 \text{ and } x = 0.$$

$$5. \text{ Test continuity of } f(x) = \begin{cases} (x-a) \cos\left(\frac{1}{x-a}\right), & x \neq a \\ 0 & , x = a \end{cases} \text{ at } x = a.$$

$$6. \text{ Test continuity of } f(x) = \begin{cases} x^2 & x < -1 \\ x & -1 \leq x < 1 \\ \frac{1}{x} & x \geq 1 \end{cases} \text{ at } x = -1 \text{ and } x = 1.$$

$$7. \text{ Test continuity of } f(x) = \begin{cases} 2^x & x \leq 1 \\ x & 1 < x \leq 4 \\ \frac{1}{x} & x > 4 \end{cases} \text{ at } x = 1 \text{ and } x = 4.$$

$$8. \text{ Test continuity of } f(x) = \begin{cases} \frac{|x-4|}{x-4}, & x \neq 4 \\ 0 & , x = 4 \end{cases} \text{ at } x = 4.$$

$$9. \text{ Test continuity of } f(x) = \begin{cases} e^{\frac{1}{x}} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$$

10. Show that $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x^2}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is continuous at $x = 0$.

Differentiability:

1. Show that $f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x^2}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is differentiable at $x = 0$.
2. Show that $f(x) = |x + 1| + |x|$ is not differentiable at $x = -1$ and $x = 0$.
3. Show that $f(x) = \begin{cases} x^2 - 1, & x \geq 1 \\ 1 - x, & x < 1 \end{cases}$ is not differentiable at $x = 1$.
4. Show that $f(x) = \begin{cases} 1 + \sin x, & 0 < x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2, & x \geq \frac{\pi}{2} \end{cases}$ is continuous and differentiable at $x = \frac{\pi}{2}$.
5. Show that $f(x) = |x|$ is continuous but not differentiable at $x = 0$.